

Al-Nassim Master Catalog

Data Dictionary & Schema Documentation

Comprehensive developer-facing documentation of the Master Catalog sheet, its 52 columns, taxonomy hierarchy, controlled vocabularies, and validation rules.

Document Version	1.0
Workbook Version	V8 (FINAL)
Total Products	2,546
Total Sheets	63
Date Generated	2026-07-06

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1. Executive Summary

This document provides a complete developer-facing specification of the Al-Nassim Master Catalog workbook — the central product information management (PIM) artifact for Al-Nassim, a Kuwait-based household goods distributor serving GCC retail channels including Amazon, Noon, Carrefour, Lulu, IKEA, Ace Hardware, and Home Centre.

The workbook contains 2,546 products across 63 sheets, with the Master Catalog sheet as the primary data table. This documentation enables another developer to fully understand the workbook structure, recreate its behavior, and integrate with it without needing to inspect the original file.

Key Facts at a Glance

Attribute	Value
Total products	2,546
Total columns in Master Catalog	52
Total sheets in workbook	63
Column groups	8 (Product Identity, Classification, Product Information, Attributes, Logistics, Commercial, SEO, Internal)
Taxonomy levels	5 (Department → Category → Subcategory → Product Family → Product Type)
Departments	2 (Houseware, Supermarket Equipment)
Categories	5
Subcategories	9
Product Families	14
Product Types	163 distinct values
Brands	2 (Al-Nassim China, Local Companies)
Languages	Bilingual (Arabic + English)
Translation audit rounds	8 (V3 through V8 + overrides)
Controlled vocabularies	15+ (Department, Category, Subcategory, Brand, Material, Color, Shape, Finish, Unit, etc.)

Document Conventions

- **Column references:** Column letters (A, B, C... AZ) refer to the physical Excel column position.

- **Mandatory/Optional:** Mandatory = must be populated; Optional = may be null.
- **Field kind:** Controlled list = value must come from a predefined dropdown. Free text = any value allowed.
- **Samples:** Sample values are drawn from the actual V8 workbook data.

2. Workbook Architecture Overview

The Al-Nassim Master Catalog workbook is a multi-sheet Excel file containing 63 sheets organized into three functional layers:

Layer 1: Core Data (1 sheet)

- Master Catalog — the primary product table with 2,546 rows × 52 columns. This is the only sheet that should be directly edited by catalog managers.

Layer 2: Reference & Taxonomy (6 sheets)

- Taxonomy Reference — defines the valid Department → Category → Subcategory hierarchy (21 entries).
- Inferred Classifications — 208 products whose classification was inferred (not direct-matched) during the V2 taxonomy project, with reviewer actions.
- Classification Summary — roll-up of the V2 classification coverage (100% achieved).
- Validation Report — taxonomy validation results.
- Translation Report — translation pipeline summary.
- Name Rewrite Report / Report V2 — summaries of the name-rewrite passes.

Layer 3: Audit & Correction Logs (56 sheets)

These sheets document the 8 rounds of translation quality assurance (V3 through V8, plus override batches). Each audit round has a paired Audit sheet (findings) and Corrections sheet (applied fixes). Key audit sheets include:

- Translation Audit V3 — the master audit log with 2,546 rows, one per product, showing action (UNCHANGED / CORRECTED / FLAG_REVIEW / APPLIED) and reason.
- Translation Audit Summary V3 — roll-up stats and top recurring issues.
- Name Rewrite Log V2 — 935 rows documenting phrase-first translation rewrites.
- Audit Change Log — 768 individual changes from the V2 audit.
- Translation Quality Corrections — 472 corrections from the quality audit.
- Cross-Validation Discrepancies V3 — 517 cross-validation findings.
- Semantic Translation Corrections — 402 semantic fixes.
- Translation QA Corrections — 1,413 corrections from the final QA pass.
- Complete Re-Audit Corr. — 474 corrections from the complete re-audit.
- Completeness Corrections — 373 corrections filling missing attributes.
- V5 Application Summary — 226 FLAG_REVIEW rows resolved.
- V6 Refinement Log — 686 rows where Product Type was upgraded from 'Other' to specific types.
- V7 Override Log — 115 rows where forbidden phrases were replaced per the Override Lexicon.

- V8 Refinement Log — 54 rows with targeted final fixes (duplicates, missing nouns, براد شاي → Tea Pot).

Sheet Naming Conventions

Pattern	Purpose	Example
* Audit	Findings from a specific QA round	Translation Quality Audit
* Corrections	Applied fixes from that round	Translation Quality Corrections
* Report	Summary of a pass	Name Rewrite Report V2
* Log	Detailed change records	Name Rewrite Log V2
* Summary	Roll-up statistics	V6 Refinement Summary
V{n} *	Version-prefixed sheet from round n	V7 Override Log
Fix{n} *	Targeted fix batch n	Fix13 Corrections

3. Master Catalog Sheet — Logical Column Groupings

The 52 columns of the Master Catalog are organized into 8 logical groups, marked by group-header cells in row 1 of the sheet. Row 2 contains the actual column headers, and data starts at row 3.

Group Overview

Group	Columns	Count	Purpose
Product Identity	A-J (1-10)	10	Unique identifiers, brand, model — answers 'which product?'
Classification	K-P (11-16)	6	Taxonomy hierarchy — answers 'where in the catalog?'
Product Information	Q-V (17-22)	6	Bilingual names and descriptions — answers 'what is it called?'
Attributes	W-AI (23-35)	13	Physical specs: color, material, capacity, dimensions — answers 'what are its properties?'
Logistics	AJ-AL (36-38)	3	Origin, sales unit, min multiples — answers 'how is it sold/shipped?'
Commercial	AM (39)	1	Pricing — answers 'how much?'
SEO	AN-AR (40-44)	5	Search-engine titles, descriptions, keywords — answers 'how is it found online?'
Internal	AS-AZ (45-52)	8	Validation status, confidence, notes, additional attributes —

			answers 'what is the data quality?'
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Visual Layout (Row 1 Group Headers)

Row 1 of the Master Catalog contains merged-cell group headers that visually span their column ranges:

A:J	K:P	Q:V	W:AI	AJ:AL	AM	AN:AR	AS:AZ	
Product Identity	Classification	Product Information	Attributes	Logistics	Com	SEO		
Internal								
(10 cols)	(6 cols)	(6 cols)	(13 cols)	(3 cols)	(1)	(5 cols)	(8 cols)	

Design Rationale

- **Ordering:** Identity first, then classification, then descriptions, then specs, then commercial, then SEO, then internal QA. This mirrors the order in which a catalog manager reads a product record.
- **Bilingual pairing:** Bilingual pairs (Name AR/EN, Color AR/EN, Material AR/EN, SEO Title AR/EN, etc.) are always adjacent to facilitate side-by-side review.
- **Spec grouping:** All dimension columns (Length, Width, Height, Diameter) are grouped together, followed by their shared Dimension Unit.
- **QA isolation:** Internal/QA columns (Validation Status, Confidence Score) are at the far right so they don't interfere with customer-facing data views.

4. Taxonomy & Hierarchy

The Master Catalog uses a 5-level taxonomy hierarchy. The first three levels (Department → Category → Subcategory) are strictly defined in the Taxonomy Reference sheet. The last two levels (Product Family → Product Type) are loosely defined by convention and actual usage.

4.1 The Five-Level Hierarchy

Level	Column	Distinct Values	Source of Truth	Enforcement
1. Department	K	2	Taxonomy Reference sheet	Strict — dropdown
2. Category	L	5	Taxonomy Reference sheet	Strict — dependent dropdown
3. Subcategory	M	9	Taxonomy Reference sheet	Strict — dependent dropdown
4. Product Family	O	14	Convention (not in Taxonomy Reference)	Loose — should be dropdown
5. Product Type	P	163	Al-Nassim Retail Lexicon	Loose — should be dependent dropdown

4.2 Complete Taxonomy Tree (from Taxonomy Reference Sheet)

The following tree shows all valid Department → Category → Subcategory combinations as defined in the Taxonomy Reference sheet. Categories marked '(no subcategory)' are leaf-level categories with no subcategory children.

```
Houseware
├── Cleaning Tools
│   ├── Surface Cleaning
│   └── Toilet & Drain
├── Home & Outdoor
│   ├── Laundry Essentials
│   ├── Picnic Collection
│   └── Storage & Packaging
├── Kitchenware
│   ├── Cooking & Utensils
│   ├── Cutlery & Cleaver
│   └── Drinkware
└── Supermarket Equipment
    ├── Accessories (no subcategory)
    └── Checkout Solutions (no subcategory)
```

- └─ Cooling Appliances
 - └─ Coolers & Chillers
 - └─ Freezers
- └─ Shelves & Stands
 - └─ Shelves
 - └─ Stands
- └─ Trolleys & Baskets
 - └─ Baskets
 - └─ Trolleys
- Warehouse Equipment
 - └─ Forklifts & Pallets
 - └─ Forklifts
 - └─ Pallets
 - └─ Heavy Duty Racking (no subcategory)
 - └─ Trolleys & Baskets
 - └─ Baskets
 - └─ Trolleys

4.3 Actual Usage in the Catalog

The following table shows how many products are actually classified under each Department → Category → Subcategory path in the current catalog:

Department	Product Count
Houseware	2543
Supermarket Equipment	3
Warehouse Equipment	3

4.4 Product Family → Product Type Mapping

Product Family (level 4) is not strictly tied to Subcategory in the current schema, but Product Type (level 5) should be constrained by Product Family. The complete mapping is in Appendix B. Here is a summary of the top 10 Product Families and their most common Product Types:

Product Family	Products	Top 3 Product Types
Buckets	415	Bucket (133), Basket (87), Box (64)
Cleaning	176	Broom (89), Brush (18), Floor Mop (15)
Cookware	378	Strainer (111), Pan (70), Cooking Pot (41)
Cutlery	242	Knife (106), Cutter (37), Scissors (28)
Drinkware	52	Cup (16), Bottle (16), Flask (15)

Food Storage	125	Container (125)
Kitchen Tools	578	Wooden Tool (56), Scale (47), Vegetable Slicer (34)
Linens	19	Towel (19)
Outdoor Cooking	80	Grill (51), Ice Tongs (11), Tool (7)
Paper Goods	4	Tissue (4)
Sets	218	Bag (107), Chair (57), Set (30)
Storage	124	Wooden Tool (35), Potato Peeler (13), Basket (11)
Tableware	94	Tray (47), Plate (36), Serving Bowl (7)
Uncategorized	41	Cooking Pot (12), Pressure Cooker (8), Food Storage Bag (5)

5. Controlled Vocabularies & Lookup Lists

The following controlled vocabularies are used in the Master Catalog. Columns marked as 'Controlled list' should only accept values from these lists. Some are currently enforced via data validation; others are convention-only and should be upgraded to dropdowns.

5.1 Brand Lookup (with Arabic + Code)

Brand (EN)	Brand AR	Brand Code	Product Count
Al-Nassim China	النسيم الصين	017	2,540
Local Companies	شركات محلية	006	1

Business rule: Brand AR and Brand Code must auto-fill via VLOOKUP when Brand is selected. The lookup table should be stored as a named range 'BrandLookup' on a hidden 'Lookups' sheet.

5.2 Department → Category → Subcategory (Dependent Dropdowns)

The complete valid combinations are defined in the Taxonomy Reference sheet. See Section 4.2 for the full tree. The dependent dropdown relationships are:

- **Level 1 → 2:** Category dropdown is filtered by the selected Department.
- **Level 2 → 3:** Subcategory dropdown is filtered by the selected Category. Some categories (Checkout Solutions, Accessories, Heavy Duty Racking) have no subcategories — Subcategory should be null.

5.3 Section Code

Section Code	Meaning	Product Count
102	Houseware section	2,537
103	Supermarket Equipment section	9

5.4 Color Lookup (with Arabic)

Color (EN)	Color AR	Product Count
Black	أسود	44
White	أبيض	33
Clear	شفاف	29
Silver	فضي	23
Gold	ذهبي	21
Red	أحمر	16
Yellow	أصفر	13

Green	أخضر	6
Blue	أزرق	6
Gray	رمادي	4
Orange	برتقالي	4
Brown	بني	3
Beige	بيج	2
Purple	بنفسجي	1
Pink	وردي	1

Business rule: Use 'Clear' for transparent products (not 'Transparent'). Use American spelling ('Gray', not 'Grey').

Color AR must auto-fill via VLOOKUP.

5.5 Material Lookup (with Arabic)

Currently 19 distinct values, but should be normalized to ~15 per the Al-Nassim Retail Lexicon:

Material (EN)	Material AR	Count	Notes
Plastic	بلاستيك	880	Correct
Stainless Steel	ستانلس ستيل	409	Correct — for استيل/ستانلس
Wood	خشب	182	Should be 'Wooden' (adjective form) per lexicon
Granite	—	50	No Arabic set — should be 'جرانيت'
Glass	زجاج	38	Correct
Metal	—	31	Generic — should be more specific
Aluminum	ألومنيوم	25	Correct
Paper	ورق	22	Correct
Iron	حديد	15	Should be 'Steel' per lexicon (حديد → Steel)
Cotton	—	12	No Arabic set

Silicone	سيليكون	11	Correct
Steel	—	11	For فولاذ/حديد — distinct from Stainless Steel
Wooden	—	8	Should merge with 'Wood' → standardize on 'Wooden'
Microfiber	—	6	No Arabic set
Sheet Steel	—	6	Should be 'Steel' or 'Saj Griddle' context
Melamine	—	3	No Arabic set — should be 'ميلامين'
Chrome	كروم	3	Correct — this is a finish, not a material
Porcelain	—	2	No Arabic set — should be 'بورسلان'
Ceramic	سيراميك	2	Correct

5.6 Unit, Capacity Unit, Weight Unit, Dimension Unit

Field	Valid Values	Current Usage
Unit (col AK)	PCS, PKT, CTN, ROLL, BUNDLE	PCS (2,499), PKT (39), CTN (6), ROLL (1), ربطه (1 — should be BUNDLE)
Capacity Unit (col AB)	L, ml, kg, g	L (309), ml (94), kg (31), g (8), KG (5 — should be kg)
Weight Unit (col AD)	kg, g	kg (41), g (6)
Dimension Unit (col AI)	cm, mm, inch	cm (575) — only cm used

5.7 Shape, Finish, Additional Information

Field	Valid Values	Current Usage
Shape (col AX)	Square, Round, Rectangular, Oval, Circular	Square (42), Round (38), Rectangular (30)

Finish (col AY)	Coated, Chrome, Black Chrome, Polished, Matte, Powder-coated	Coated (9), Black Chrome (2), Chrome (1)
Additional Information (col AZ)	with Lid, Glass Lid, with Wheels, Foldable, with Handle	'with Lid' (50), 'Glass Lid' (29)

5.8 Validation Status & Confidence Score

Field	Valid Values	Current Usage
Validation Status (col AT)	Validated, Pending Review, Draft, Rejected	Validated (2,546 — all rows)
Confidence Score (col AU)	Integer 0–100	97 (1,337), 94 (978), 82 (117), 79 (44), 92 (35), 100 (28), 89 (2), 77 (2), 95 (2), 85 (1)

Confidence score bands: 100 = perfect match; 97 = high-confidence lexicon match; 94 = standard lexicon match; 82–89 = inferred (needs review); 77–79 = low confidence.

5.9 Country of Origin

Currently only 'China' is in use (876 rows). The full ISO 3166 country list should be the lookup source for this dropdown. Future expected values: India, Turkey, UAE, Germany, Italy, Vietnam.

6. Column-by-Column Data Dictionary

This section documents every column in the Master Catalog sheet. Columns are presented in their logical group order (not physical column order). Each column entry covers all 10 documentation attributes: purpose, data type, mandatory, validation, field kind, valid options, dependencies, defaults, duplicates, and business rules.

6.1 Product Identity Group (Columns A–J)

Source Row (Column A, #1)

Distinct values: 2,546 | Non-null: 2,546 | Null: 0

1. Purpose	Internal row-number reference pointing back to the original source workbook from which this catalog was migrated. Used for traceability during audits and data reconciliation.
2. Data Type	Integer (positive)
3. Mandatory	Yes
4. Validation Rules	Must be a positive integer. Range: 1–2548. Must be unique across all rows.
5. Field Kind	Auto-generated integer
6. Valid Options	N/A — auto-generated
7. Dependencies	None
8. Default Values	Auto-assigned sequentially when a new row is created.
9. Duplicates	No — must be unique. Acts as the primary internal row key.
10. Business Rules	Never edited manually. Used by audit sheets (Translation Audit V3, V4, V5, V6, V7, V8) to reference back to the Master Catalog row. Sample values: 59, 66, 73.

Sample values: '59' | '66' | '295'

Product ID (Column B, #2)

Distinct values: 2,546 | Non-null: 2,546 | Null: 0

1. Purpose	The globally unique product identifier used by the ERP/PIM system. This is the canonical business key for
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	the product and never changes once assigned.
2. Data Type	String (numeric, typically 12 digits)
3. Mandatory	Yes
4. Validation Rules	Must be a non-empty string. In practice always numeric, 12 digits. Must be unique.
5. Field Kind	Free text (but should be numeric)
6. Valid Options	N/A — assigned by ERP
7. Dependencies	None
8. Default Values	Assigned by the ERP system at product creation time.
9. Duplicates	No — must be globally unique.
10. Business Rules	This is the primary key for integration with ERP, e-commerce, and warehouse systems. Never reused even after a product is retired. Sample values: 102000000126, 102000000135.

Sample values: '102000000126' | '102000000135' | '102000000433'

SKU (Column C, #3)

Distinct values: 2,546 | Non-null: 2,546 | Null: 0

1. Purpose	Stock Keeping Unit — the warehouse-facing identifier used for inventory management, picking, and stock movements. Often equals Product ID in this catalog but is conceptually distinct (a SKU can differ if the same product is stored in multiple warehouses).
2. Data Type	String (numeric, 12 digits)
3. Mandatory	Yes
4. Validation Rules	Must be non-empty. In this catalog it mirrors Product ID. Must be unique.
5. Field Kind	Free text
6. Valid Options	N/A
7. Dependencies	Usually equals Product ID
8. Default Values	Defaults to Product ID unless overridden for multi-warehouse scenarios.

9. Duplicates	No — must be unique within a warehouse.
10. Business Rules	Used by WMS (Warehouse Management System) for bin/pick logic. Sample values: 102000000126, 102000000135.

Sample values: '102000000126' | '102000000135' | '102000000433'

ND Number (Column D, #4)

Distinct values: 2,344 | Non-null: 2,391 | Null: 155

1. Purpose	Al-Nassim's internal product number, used as the primary human-readable reference in retail catalogs, price lists, and the Arabic name string. This is the number customers see on invoices and shelf labels.
2. Data Type	String (alphanumeric, format: ND-XXXX or ND-XXXX-XXXX[-XXXX])
3. Mandatory	Optional (155 rows have null — typically for locally-sourced products without an ND code)
4. Validation Rules	Format: ND- followed by 4 digits, optionally followed by additional segments (model/variant codes). Regex: <code>^ND-\d{4}(-[A-Z0-9-]+)?\$</code>
5. Field Kind	Free text with format constraint
6. Valid Options	N/A — assigned by Al-Nassim
7. Dependencies	None
8. Default Values	None — assigned at product onboarding.
9. Duplicates	Duplicates allowed in rare cases where the same ND number covers multiple color/size variants. 2,344 distinct values across 2,546 rows.
10. Business Rules	Appears in the Name EN string (e.g., 'Al-Nassim ND-5921 Stainless Steel Cooking Pot'). Used for SEO and customer-facing lookups. Sample values: ND-5921, ND-5091, ND-36029-100.

Sample values: 'ND-1220' | 'ND-1030-K812' | 'ND-1031-135E'

Barcode (Column E, #5)

Distinct values: 2,546 | Non-null: 2,546 | Null: 0

1. Purpose	EAN-13 / UPC-A barcode printed on the product packaging. Used for POS scanning, inventory receipt, and e-commerce marketplace listings (Amazon, Noon, Carrefour).
2. Data Type	String (numeric, 13 digits for EAN-13)
3. Mandatory	Yes
4. Validation Rules	Must be 13 digits (EAN-13) or 12 digits (UPC-A). Must pass EAN/UPC checksum validation. Must be unique.
5. Field Kind	Free text (numeric only)
6. Valid Options	N/A — assigned by GS1
7. Dependencies	None
8. Default Values	None — obtained from GS1 prefix allocated to the brand.
9. Duplicates	No — must be globally unique (GS1 standard).
10. Business Rules	Required for listing on Amazon, Noon, Carrefour, Lulu. Cannot be changed once printed on packaging. Sample values: 4701512038142, 4701512038166.

Sample values: '4701512038142' | '4701512038166' | '6921080164069'

Legacy Code (Column F, #6)

Distinct values: 411 | Non-null: 411 | Null: 2,135

1. Purpose	Product code from the previous/legacy ERP system, retained for historical lookup and reconciliation during migration. Will be deprecated once migration is complete.
2. Data Type	String (alphanumeric)
3. Mandatory	Optional (2,135 rows have null — these products have no legacy code)
4. Validation Rules	Free text. No format constraint.
5. Field Kind	Free text
6. Valid Options	N/A

7. Dependencies	None
8. Default Values	None
9. Duplicates	Duplicates allowed (same legacy code can map to multiple new SKUs if the legacy system merged variants).
10. Business Rules	Read-only after migration. Used only for backward-compatibility lookups. Sample values: (various — 411 distinct values).

Sample values: '8851246900010' | '6924545212541' | '1020000011683'

Brand (Column G, #7)

Distinct values: 2 | Non-null: 2,541 | Null: 5

1. Purpose	The brand name in English, as displayed on the product packaging and e-commerce listings. Identifies the manufacturer or importer.
2. Data Type	String
3. Mandatory	Yes (5 rows have null — these should be populated)
4. Validation Rules	Must come from the controlled Brand list (currently 2 values).
5. Field Kind	Controlled list (dropdown)
6. Valid Options	Al-Nassim China Local Companies
7. Dependencies	Determines Brand AR and Brand Code (auto-filled when Brand is selected).
8. Default Values	Al-Nassim China (the default for 99.8% of products).
9. Duplicates	Duplicates expected — many products share the same brand.
10. Business Rules	When a new brand is onboarded, add it to the Brand lookup list first, then update Brand AR and Brand Code accordingly. Sample values: Al-Nassim China, Local Companies.

Sample values: 'Al-Nassim China' | 'Al-Nassim China' | 'Al-Nassim China'

Brand AR (Column H, #8)

Distinct values: 2 | Non-null: 2,541 | Null: 5

1. Purpose	The brand name in Arabic, for Arabic-language catalog pages, invoices, and shelf labels in Kuwait/GCC markets.
2. Data Type	String (Arabic)
3. Mandatory	Yes (same 5 null rows as Brand)
4. Validation Rules	Must correspond to the Brand selection (1:1 mapping).
5. Field Kind	Controlled list (auto-derived from Brand)
6. Valid Options	النسيم الصين (for Al-Nassim China) شركات محلية (for Local Companies)
7. Dependencies	Directly dependent on Brand — should auto-fill when Brand is selected.
8. Default Values	النسيم الصين
9. Duplicates	Duplicates expected.
10. Business Rules	Never edit manually — should be auto-populated via VLOOKUP from the Brand lookup table. Sample values: النسيم الصين, شركات محلية.

Sample values: 'النسيم الصين' | 'النسيم الصين' | 'النسيم الصين'

Brand Code (Column I, #9)

Distinct values: 2 | Non-null: 2,541 | Null: 5

1. Purpose	Short numeric brand identifier used by the ERP for compact references (e.g., in barcode prefixes, invoice line items).
2. Data Type	String (3-digit numeric)
3. Mandatory	Yes (same 5 null rows)
4. Validation Rules	Must be a 3-digit string. Must correspond to the Brand selection.
5. Field Kind	Controlled list (auto-derived from Brand)
6. Valid Options	017 (for Al-Nassim China) 006 (for Local Companies)
7. Dependencies	Directly dependent on Brand — auto-fills when Brand

	is selected.
8. Default Values	017
9. Duplicates	Duplicates expected.
10. Business Rules	Never edit manually — auto-populated via VLOOKUP. Sample values: 017, 006.

Sample values: '017' | '017' | '017'

Model (Column J, #10)

Distinct values: 2,443 | Non-null: 2,510 | Null: 36

1. Purpose	The manufacturer's model / variant code, as printed on the product or its packaging. Distinguishes variants within the same ND number (e.g., different colors, sizes).
2. Data Type	String (alphanumeric)
3. Mandatory	Optional (36 rows have null)
4. Validation Rules	Free text. Typically uppercase alphanumeric with hyphens (e.g., MJ-35, BC-06, HY-63-6).
5. Field Kind	Free text
6. Valid Options	N/A — assigned by manufacturer
7. Dependencies	Often appears in the Name EN string after the ND number.
8. Default Values	None
9. Duplicates	Duplicates allowed (same model can be used across ND numbers).
10. Business Rules	Used for variant matching when receiving goods. Should be preserved exactly as provided by the supplier. Sample values: MJ-35, BC-06, HY-63-6, FEM7525.

Sample values: 'SH422' | 'SH453' | 'SH0462'

6.2 Classification Group (Columns K–P)

Department (Column K, #11)

Distinct values: 2 | Non-null: 2,546 | Null: 0

1. Purpose	Top level of the product taxonomy. Identifies the major business division the product belongs to.
2. Data Type	String
3. Mandatory	Yes
4. Validation Rules	Must come from the controlled Department list. Currently 2 values.
5. Field Kind	Controlled list (dropdown)
6. Valid Options	Houseware Supermarket Equipment Warehouse Equipment (defined in Taxonomy Reference sheet; only first two are currently in use)
7. Dependencies	Determines the available Categories (dependent dropdown).
8. Default Values	Houseware
9. Duplicates	Duplicates expected.
10. Business Rules	Never change a product's Department once set — it cascades to Category, Subcategory, and reporting. New departments require Taxonomy Reference sheet update first. Sample values: Houseware (2,543 rows), Supermarket Equipment (3 rows).

Sample values: 'Houseware' | 'Houseware' | 'Houseware'

Category (Column L, #12)

Distinct values: 5 | Non-null: 2,546 | Null: 0

1. Purpose	Second level of the taxonomy. Groups related subcategories within a Department.
2. Data Type	String
3. Mandatory	Yes
4. Validation Rules	Must come from the controlled Category list AND must be valid under the selected Department (dependent

	dropdown).
5. Field Kind	Controlled list (dependent dropdown — depends on Department)
6. Valid Options	Under Houseware: Kitchenware Cleaning Tools Home & Outdoor. Under Supermarket Equipment: Cooling Appliances Trolleys & Baskets Shelves & Stands Checkout Solutions Accessories. Under Warehouse Equipment: Trolleys & Baskets Forklifts & Pallets Heavy Duty Racking.
7. Dependencies	Depends on Department. The valid Category list changes based on Department selection.
8. Default Values	Kitchenware (most common)
9. Duplicates	Duplicates expected.
10. Business Rules	Changing Department without updating Category is forbidden. Sample values: Kitchenware (1,664), Cleaning Tools (499), Home & Outdoor (380).

Sample values: 'Kitchenware' | 'Kitchenware' | 'Cleaning Tools'

Subcategory (Column M, #13)

Distinct values: 9 | Non-null: 2,544 | Null: 2

1. Purpose	Third level of the taxonomy. Provides granular classification for navigation, filtering, and reporting.
2. Data Type	String
3. Mandatory	Yes (2 rows have null — these are Checkout Solutions / Accessories which have no subcategory)
4. Validation Rules	Must come from the controlled Subcategory list AND must be valid under the selected Category (dependent dropdown).
5. Field Kind	Controlled list (dependent dropdown — depends on Category)
6. Valid Options	Under Houseware/Kitchenware: Cooking & Utensils Cutlery & Cleaver Drinkware. Under Houseware/Cleaning Tools: Surface Cleaning Toilet & Drain. Under Houseware/Home & Outdoor: Laundry

	Essentials Picnic Collection Storage & Packaging. Under Supermarket Equipment/Cooling Appliances: Coolers & Chillers Freezers. Under Supermarket Equipment/Trolleys & Baskets: Baskets Trolleys. Under Supermarket Equipment/Shelves & Stands: Shelves Stands. Under Supermarket Equipment/Checkout Solutions: (none). Under Supermarket Equipment/Accessories: (none). Under Warehouse Equipment/Trolleys & Baskets: Baskets Trolleys. Under Warehouse Equipment/Forklifts & Pallets: Forklifts Pallets. Under Warehouse Equipment/Heavy Duty Racking: (none).
7. Dependencies	Depends on Category (which depends on Department).
8. Default Values	Cooking & Utensils
9. Duplicates	Duplicates expected.
10. Business Rules	Some categories (Checkout Solutions, Accessories, Heavy Duty Racking) have no subcategories — Subcategory should be null for these. Sample values: Cooking & Utensils (1,263), Surface Cleaning (478), Cutlery & Cleaver (366).

Sample values: 'Cooking & Utensils' | 'Cooking & Utensils' | 'Surface Cleaning'

Section Code (Column N, #14)

Distinct values: 2 | Non-null: 2,546 | Null: 0

1. Purpose	Legacy numeric code (3-digit) identifying the retail section/aisle where the product is merchandised in physical stores.
2. Data Type	String (3-digit numeric)
3. Mandatory	Yes
4. Validation Rules	Must be a 3-digit string. Currently only 2 values in use.
5. Field Kind	Controlled list (dropdown)
6. Valid Options	102 (Houseware section) 103 (Supermarket Equipment section)
7. Dependencies	Should correspond to Department (102 = Houseware,

	103 = Supermarket Equipment).
8. Default Values	102
9. Duplicates	Duplicates expected.
10. Business Rules	Used by store-layout software and planogram tools. Sample values: 102 (2,537 rows), 103 (9 rows).

Sample values: '102' | '102' | '102'

Product Family (Column O, #15)

Distinct values: 14 | Non-null: 2,546 | Null: 0

1. Purpose	Logical grouping of related Product Types. Used for reporting, catalog navigation, and cross-sell rules. More granular than Subcategory but broader than Product Type.
2. Data Type	String
3. Mandatory	Yes
4. Validation Rules	Should come from the controlled Product Family list (14 values currently in use).
5. Field Kind	Controlled list (dropdown) — currently free-text in practice
6. Valid Options	Kitchen Tools Cookware Cutlery Cleaning Food Storage Storage Tableware Outdoor Cooking Drinkware Sets Buckets Linens Paper Goods Uncategorized
7. Dependencies	Loosely correlated with Subcategory but not strictly enforced (a Subcategory can contain multiple Product Families).
8. Default Values	None — must be selected.
9. Duplicates	Duplicates expected.
10. Business Rules	Recommendation: enforce as a dependent dropdown based on Subcategory. Currently 14 distinct values. Sample: Kitchen Tools (578), Buckets (415), Cookware (378).

Sample values: 'Cookware' | 'Cookware' | 'Kitchen Tools'

Product Type (Column P, #16)

Distinct values: 163 | Non-null: 2,546 | Null: 0

1. Purpose	The specific commercial product name — the most granular classification level. This is the noun that appears in the Name EN string (e.g., 'Cooking Pot', 'Kitchen Knife', 'BBQ Skewer').
2. Data Type	String
3. Mandatory	Yes
4. Validation Rules	Should come from the controlled Product Type list. Currently 163 distinct values (high cardinality).
5. Field Kind	Controlled list (dropdown) — should be dependent on Product Family
6. Valid Options	163 distinct values. Top 20 by frequency: Strainer (158), Container (134), Bucket (133), Wooden Tool (111), Knife (109), Cutter (107), Bag (107), Basket (98), Broom (90), Box (76), Pan (70), Tray (67), Chair (57), Set (56), Cooking Pot (53), Grill (51), Scale (47), Flask (45), Plate (36), Vegetable Slicer (35).
7. Dependencies	Should depend on Product Family (each Family has a set of valid Product Types). See Product Family → Product Types mapping in the appendix.
8. Default Values	None — must be selected.
9. Duplicates	Duplicates expected (many products share the same Product Type).
10. Business Rules	Critical for catalog search and filtering. Must follow the Al-Nassim Retail Lexicon (no generic terms like 'Other', 'Machine', 'Tool'). Sample: 'Cooking Pot', 'Frying Pan', 'Toilet Brush'.

Sample values: 'Pan' | 'Pan' | 'Sprayer'

6.3 Product Information Group (Columns Q–V)

Name AR (Column Q, #17)

Distinct values: 2,535 | Non-null: 2,546 | Null: 0

1. Purpose	The full product name in Arabic, as provided by the supplier. This is the source-of-truth name from which Name EN is translated.
2. Data Type	String (Arabic)
3. Mandatory	Yes
4. Validation Rules	Must be non-empty Arabic text. Should contain the brand, model/ND number, attributes (material, size, color, quantity), and the product noun.
5. Field Kind	Free text (Arabic)
6. Valid Options	N/A
7. Dependencies	Source for Name EN translation.
8. Default Values	None
9. Duplicates	Duplicates rare (2,535 distinct values out of 2,546 rows — 11 duplicate names exist for variant products).
10. Business Rules	Never modified by the translation pipeline — it is the input. Sample: 'كسرولة حليب 14 سم Olina'.

Sample values: 'كسرولة حليب 14 سم Olina' | 'كسرولة حليب 16 سم Olina' | 'مطارة ماء اطفال Al-nassim SH422'

Name EN (Column R, #18)

Distinct values: 2,519 | Non-null: 2,546 | Null: 0

1. Purpose	The full product name in English, translated from Name AR using the Al-Nassim Retail Lexicon. This is the customer-facing title shown on e-commerce sites, invoices, and price lists.
2. Data Type	String (English)
3. Mandatory	Yes
4. Validation Rules	Must follow the commercial naming format: Brand → ND Number → Quantity → Material → Handle → Shape → Product → Accessory → Capacity →

	Dimensions. Must use lexicon-approved terms (no 'Bin', 'Casserole', 'Sheet', 'Flask' for براد شاي, etc.).
5. Field Kind	Free text (English) — derived from Name AR via translation
6. Valid Options	N/A — but must comply with the Al-Nassim Retail Lexicon
7. Dependencies	Derived from Name AR. Must preserve all attributes (material, dimensions, capacity, quantity, color, ND number).
8. Default Values	None
9. Duplicates	Duplicates rare (2,519 distinct values).
10. Business Rules	Subject to V3–V8 translation audits. Must not contain forbidden phrases (Food Container, Sweeper, Shovel, etc.). Must preserve product identity (same physical product as Name AR). Sample: 'Al-Nassim ND-5921 Stainless Steel Cooking Pot with Glass Lid, 26 cm'.

Sample values: 'Al-Nassim 14 cm Milk Saucepan' | 'Al-Nassim 16 cm Milk Saucepan' | 'Al-Nassim SH422 Children's Water Bottle'

Short Desc AR (Column S, #19)

Distinct values: 2,535 | Non-null: 2,546 | Null: 0

1. Purpose	Short product description in Arabic (1–2 sentences). Used in product listing cards, search results, and mobile views where space is limited.
2. Data Type	String (Arabic)
3. Mandatory	Yes (defaults to Name AR if not explicitly written)
4. Validation Rules	Free text, max ~150 characters recommended.
5. Field Kind	Free text (Arabic)
6. Valid Options	N/A
7. Dependencies	Often mirrors Name AR.
8. Default Values	Defaults to Name AR value.
9. Duplicates	Duplicates common (2,535 distinct — mirrors Name

	AR).
10. Business Rules	Displayed in product cards on Arabic storefronts. Sample: 'كسرولة حليب 14 سم Olina'.

Sample values: 'كسرولة حليب 14 سم Olina' | 'كسرولة حليب 16 سم Olina' | 'مطارة ماء اطفال Al-nassim SH422'

Short Desc EN (Column T, #20)

Distinct values: 2,519 | Non-null: 2,546 | Null: 0

1. Purpose	Short product description in English (1–2 sentences). Used in product listing cards, search results, and mobile views.
2. Data Type	String (English)
3. Mandatory	Yes (defaults to Name EN)
4. Validation Rules	Free text, max ~150 characters recommended.
5. Field Kind	Free text (English)
6. Valid Options	N/A
7. Dependencies	Often mirrors Name EN.
8. Default Values	Defaults to Name EN value.
9. Duplicates	Duplicates common (2,519 distinct).
10. Business Rules	Displayed in product cards on English storefronts. Sample: 'Al-Nassim 14 cm Milk Saucepan'.

Sample values: 'Al-Nassim 14 cm Milk Saucepan' | 'Al-Nassim 16 cm Milk Saucepan' | 'Al-Nassim SH422 Children's Water Bottle'

Long Desc AR (Column U, #21)

Distinct values: 12 | Non-null: 673 | Null: 1,873

1. Purpose	Long-form product description in Arabic for the product detail page. Used for SEO and customer education.
2. Data Type	String (Arabic)
3. Mandatory	Optional (1,873 rows have null — long descriptions are

	only written for premium/featured products)
4. Validation Rules	Free text, max ~500 characters recommended.
5. Field Kind	Free text (Arabic)
6. Valid Options	N/A
7. Dependencies	None
8. Default Values	None
9. Duplicates	Duplicates common (only 12 distinct values — mostly boilerplate 'النسيم الصين', Other').
10. Business Rules	Only populated for featured products. Sample: 'النسيم الصين', Other'.

Sample values: ميزان الكتروني

ماركة : CONSTANT

444 : F-ND-1035 موديل

صناعة : الصين

الإرتفاع : 120 سم

النوع :- مناشف ميكروفايبر | 'MAX 30KG' : الوزن

Type: Microfiber towels

العدد:- 25 قطعة

Quantity:- 25 pieces

سم 30xالمقاس :- 30

Size:- 30x30 cm

رقم الصنف :- Al-nassim ND-5001

Item number: Al-nassim ND-5001

شد الكرتون :- 32 باكيت

النوع :- مناشف ميكروفايبر | '32 packets' : Carton packing

Type: Microfiber towels

العدد:- 50 قطعة

Quantity:- 50 pieces

سم 30xالمقاس :- 30

Size:- 30x30 cm

رقم الصنف :- Al-nassim ND-5002

Item number: Al-nassim ND-5002

شد الكرتون :- 18 باكيت

Carton packing: 18 packets'

Long Desc EN (Column V, #22)

Distinct values: 2,259 | Non-null: 2,546 | Null: 0

1. Purpose	Long-form product description in English for the product detail page. Used for SEO and customer education.
2. Data Type	String (English)
3. Mandatory	Yes (no nulls — defaults to a generated sentence based on Brand + Name EN)
4. Validation Rules	Free text.
5. Field Kind	Free text (English)
6. Valid Options	N/A
7. Dependencies	Auto-generated from Brand + Name EN if not explicitly written.
8. Default Values	Defaults to '{Brand}. {Name EN}.' format.
9. Duplicates	Duplicates common (2,259 distinct values).
10. Business Rules	Should be expanded for SEO on key products. Sample: 'Al-Nassim China. Al-Nassim 14 cm Milk Saucepan.'

Sample values: 'Al-Nassim China. Al-Nassim 14 cm Milk Saucepan.' | 'Al-Nassim China. Al-Nassim 16 cm Milk Saucepan.' | 'Al-Nassim China. Al-Nassim SH422 Children's Water Bottle Product Family: Garden Tools.'

6.4 Attributes Group (Columns W–AI)

Color (Column W, #23)

Distinct values: 15 | Non-null: 206 | Null: 2,340

1. Purpose	The product's color in English, for filtering and variant selection on e-commerce sites.
2. Data Type	String
3. Mandatory	Optional (2,340 rows have null — color only specified when the product comes in distinct colors)
4. Validation Rules	Should come from the controlled Color list (15 values).
5. Field Kind	Controlled list (dropdown)
6. Valid Options	Black White Clear Silver Gold Red Yellow Green Blue Gray Orange Brown Beige Purple

	Pink
7. Dependencies	Must correspond to Color AR (1:1 mapping).
8. Default Values	None
9. Duplicates	Duplicates expected.
10. Business Rules	Use 'Clear' for transparent products (not 'Transparent'). Use 'Gray' (American spelling). Sample: Black (44), White (33), Clear (29).

Sample values: 'Red' | 'Red' | 'Red'

Color AR (Column X, #24)

Distinct values: 15 | Non-null: 206 | Null: 2,340

1. Purpose	The product's color in Arabic, for Arabic-language storefronts.
2. Data Type	String (Arabic)
3. Mandatory	Optional (same 2,340 nulls as Color)
4. Validation Rules	Must correspond to Color selection (1:1 mapping).
5. Field Kind	Controlled list (auto-derived from Color)
6. Valid Options	أسود (Black) أبيض (White) شفاف (Clear) فضي (Silver) ذهبي (Gold) أحمر (Red) أصفر (Yellow) أخضر (Green) أزرق (Blue) رمادي (Gray) برتقالي (Orange) بني (Brown) بيج (Beige) بنفسجي (Purple) وردي (Pink)
7. Dependencies	Directly dependent on Color — should auto-fill via VLOOKUP.
8. Default Values	None
9. Duplicates	Duplicates expected.
10. Business Rules	Never edit manually — auto-populated from Color lookup. Sample: شفاف, أبيض, أسود.

Sample values: 'أحمر' | 'أحمر' | 'أحمر'

Material (Column Y, #25)

Distinct values: 19 | Non-null: 1,716 | Null: 830

1. Purpose	The primary material of the product in English, for filtering, care instructions, and SEO.
2. Data Type	String
3. Mandatory	Optional (830 rows have null — material only specified when material is a key buying factor)
4. Validation Rules	Should come from the controlled Material list (19 values currently in use, but should be normalized to ~15).
5. Field Kind	Controlled list (dropdown)
6. Valid Options	Plastic Stainless Steel Wood Granite Glass Metal Aluminum Paper Iron Cotton Silicone Steel Wooden Microfiber Sheet Steel Melamine Chrome Porcelain Ceramic. NOTE: 'Wood' and 'Wooden' should be normalized to 'Wooden' (adjective form per lexicon). 'Steel' and 'Sheet Steel' should be normalized to 'Stainless Steel' or 'Steel' respectively.
7. Dependencies	Must correspond to Material AR.
8. Default Values	None
9. Duplicates	Duplicates expected.
10. Business Rules	Per the Al-Nassim Retail Lexicon: استيل → 'Stainless Steel' (never 'Steel'), خشب → 'Wooden' (never 'Wood' as noun), حديد → 'Steel' (never 'Iron'), فولاذ → 'Steel'. Sample: Plastic (880), Stainless Steel (409), Wood (182).

Sample values: 'Plastic' | 'Plastic' | 'Plastic'

Material AR (Column Z, #26)

Distinct values: 10 | Non-null: 1,637 | Null: 909

1. Purpose	The product's material in Arabic, for Arabic-language storefronts.
2. Data Type	String (Arabic)
3. Mandatory	Optional (909 nulls)
4. Validation Rules	Must correspond to Material selection.

5. Field Kind	Controlled list (auto-derived from Material)
6. Valid Options	بلاستيك (Plastic) ستانلس ستيل (Stainless Steel) خشب (Wood/Wooden) زجاج (Glass) حديد (Iron/Steel) ألومنيوم (Aluminum) ورق (Paper) سيليكون (Silicone) سيراميك (Ceramic) كروم (Chrome)
7. Dependencies	Directly dependent on Material — auto-fill via VLOOKUP.
8. Default Values	None
9. Duplicates	Duplicates expected.
10. Business Rules	Never edit manually. Sample: بلاستيك, ستانلس ستيل, خشب.

Sample values: 'بلاستيك' | 'ستانلس ستيل' | 'خشب'

Capacity (Column AA, #27)

Distinct values: 144 | Non-null: 447 | Null: 2,099

1. Purpose	The numeric capacity/volume/weight of the product (e.g., 1.5 liters, 500 ml, 5 kg). Used for filtering and SEO.
2. Data Type	Number (decimal)
3. Mandatory	Optional (2,099 rows have null — capacity only relevant for containers, cookware, etc.)
4. Validation Rules	Must be a positive number. Must be accompanied by Capacity Unit.
5. Field Kind	Numeric
6. Valid Options	N/A — any positive number
7. Dependencies	Paired with Capacity Unit.
8. Default Values	None
9. Duplicates	Duplicates expected.
10. Business Rules	Must appear in the Name EN string (e.g., '1.5 L'). Sample values: 1.5, 2, 5, 500, 14.

Sample values: '1.25' | '2.8' | '1.5'

Capacity Unit (Column AB, #28)

Distinct values: 5 | Non-null: 447 | Null: 2,099

1. Purpose	The unit of measurement for the Capacity field.
2. Data Type	String
3. Mandatory	Optional (same 2,099 nulls as Capacity)
4. Validation Rules	Must come from the controlled unit list. Currently 5 values (should be normalized to 4 — 'KG' and 'kg' are duplicates).
5. Field Kind	Controlled list (dropdown)
6. Valid Options	L (liters) ml (milliliters) kg (kilograms) g (grams). NOTE: 'KG' should be normalized to 'kg'.
7. Dependencies	Paired with Capacity.
8. Default Values	None
9. Duplicates	Duplicates expected.
10. Business Rules	Use lowercase 'ml', 'kg', 'g'. Use uppercase 'L' for liters (per retail convention). Sample: L (309), ml (94), kg (31).

Sample values: 'L' | 'L' | 'L'

Weight (Column AC, #29)

Distinct values: 16 | Non-null: 47 | Null: 2,499

1. Purpose	The product's weight (excluding packaging), for shipping calculations and logistics.
2. Data Type	Number (decimal)
3. Mandatory	Optional (2,499 rows have null — weight only measured for heavier products)
4. Validation Rules	Must be a positive number. Must be accompanied by Weight Unit.
5. Field Kind	Numeric
6. Valid Options	N/A
7. Dependencies	Paired with Weight Unit.

8. Default Values	None
9. Duplicates	Duplicates expected.
10. Business Rules	Used by shipping cost calculators. Sample values: 0.3, 0.9, 1.2, 2.5.

Sample values: '5' | '30' | '30'

Weight Unit (Column AD, #30)

Distinct values: 2 | Non-null: 47 | Null: 2,499

1. Purpose	The unit for the Weight field.
2. Data Type	String
3. Mandatory	Optional (same nulls as Weight)
4. Validation Rules	Must come from controlled list.
5. Field Kind	Controlled list (dropdown)
6. Valid Options	kg g
7. Dependencies	Paired with Weight.
8. Default Values	kg
9. Duplicates	Duplicates expected.
10. Business Rules	Use lowercase. Sample: kg (41), g (6).

Sample values: 'kg' | 'kg' | 'kg'

Length (Column AE, #31)

Distinct values: 87 | Non-null: 559 | Null: 1,987

1. Purpose	The product's length in the unit specified by Dimension Unit. For cookware, this is typically the diameter.
2. Data Type	Number (decimal)
3. Mandatory	Optional (1,987 rows have null — dimensions only specified for size-relevant products)
4. Validation Rules	Must be a positive number. Should be accompanied by Width or Diameter for 2D/3D products.

5. Field Kind	Numeric
6. Valid Options	N/A
7. Dependencies	Paired with Dimension Unit.
8. Default Values	None
9. Duplicates	Duplicates expected.
10. Business Rules	For round products (pans, pots), use Diameter instead. Sample values: 14, 16, 20, 24, 26, 30, 32.

Sample values: '14' | '16' | '36'

Width (Column AF, #32)

Distinct values: 48 | Non-null: 102 | Null: 2,444

1. Purpose	The product's width in Dimension Unit.
2. Data Type	Number (decimal)
3. Mandatory	Optional (2,444 nulls)
4. Validation Rules	Must be a positive number.
5. Field Kind	Numeric
6. Valid Options	N/A
7. Dependencies	Paired with Length and Dimension Unit.
8. Default Values	None
9. Duplicates	Duplicates expected.
10. Business Rules	Only populated for rectangular/square products. Sample values: 7, 10, 15, 20.

Sample values: '30' | '35' | '40'

Height (Column AG, #33)

Distinct values: 0 | Non-null: 0 | Null: 2,546

1. Purpose	The product's height in Dimension Unit.
2. Data Type	Number (decimal)
3. Mandatory	Optional (all 2,546 rows have null — height is not currently captured)

4. Validation Rules	Must be a positive number.
5. Field Kind	Numeric
6. Valid Options	N/A
7. Dependencies	Paired with Dimension Unit.
8. Default Values	None
9. Duplicates	N/A (column is empty).
10. Business Rules	Recommendation: populate for tall products (mops, brooms, chairs) to improve shipping calculations. Currently unused.

Diameter (Column AH, #34)

Distinct values: 11 | Non-null: 16 | Null: 2,530

1. Purpose	The diameter of round products (pans, pots, lids) in Dimension Unit.
2. Data Type	Number (decimal)
3. Mandatory	Optional (2,530 nulls — only relevant for round products)
4. Validation Rules	Must be a positive number. Should not coexist with Length/Width (use one or the other).
5. Field Kind	Numeric
6. Valid Options	N/A
7. Dependencies	Paired with Dimension Unit.
8. Default Values	None
9. Duplicates	Duplicates expected.
10. Business Rules	Used for cookware sizing (e.g., '26 cm Frying Pan'). Sample values: 14, 16, 20, 24, 26, 28, 30, 32.

Sample values: '16' | '22' | '28'

Dimension Unit (Column AI, #35)

Distinct values: 1 | Non-null: 575 | Null: 1,971

1. Purpose	The unit for Length, Width, Height, and Diameter
-------------------	--

	fields.
2. Data Type	String
3. Mandatory	Optional (1,971 nulls — only populated when a dimension is set)
4. Validation Rules	Must come from controlled list. Currently only 1 value in use.
5. Field Kind	Controlled list (dropdown)
6. Valid Options	cm (centimeters). Future options: mm, inch.
7. Dependencies	Paired with any dimension field.
8. Default Values	cm
9. Duplicates	Duplicates expected.
10. Business Rules	Always 'cm' in this catalog. Use lowercase. Sample: cm (575 rows).

Sample values: 'cm' | 'cm' | 'cm'

6.5 Logistics Group (Columns AJ–AL)

Country of Origin (Column AJ, #36)

Distinct values: 1 | Non-null: 876 | Null: 1,670

1. Purpose	The country where the product was manufactured, for customs, labeling, and customer information.
2. Data Type	String
3. Mandatory	Optional (1,670 nulls — only specified for branded products)
4. Validation Rules	Should come from the ISO 3166 country list. Currently only 1 value in use.
5. Field Kind	Controlled list (dropdown)
6. Valid Options	China. Future options: India, Turkey, UAE, Germany, Italy, etc.
7. Dependencies	None
8. Default Values	China
9. Duplicates	Duplicates expected.

10. Business Rules	Required for customs declarations in Kuwait/GCC. Sample: China (876 rows).

Sample values: 'China' | 'China' | 'China'

Unit (Column AK, #37)

Distinct values: 5 | Non-null: 2,546 | Null: 0

1. Purpose	The unit of sale — how the product is counted when ordered/invoiced.
2. Data Type	String
3. Mandatory	Yes
4. Validation Rules	Must come from controlled list. Currently 5 values (1 is Arabic — should be normalized).
5. Field Kind	Controlled list (dropdown)
6. Valid Options	PCS (pieces) PKT (packet) CTN (carton) ROLL ربطه (Arabic for 'bundle' — should be normalized to 'BUNDLE').
7. Dependencies	None
8. Default Values	PCS
9. Duplicates	Duplicates expected.
10. Business Rules	PCS = individual piece, PKT = multi-piece packet, CTN = full carton, ROLL = roll of tape/table. Sample: PCS (2,499), PKT (39), CTN (6).

Sample values: 'PCS' | 'PCS' | 'PCS'

Min Sales Multiples (Column AL, #38)

Distinct values: 1 | Non-null: 2,546 | Null: 0

1. Purpose	Indicates whether the product must be sold in multiples of a base quantity (e.g., sold by the carton of 12).
2. Data Type	String (boolean-like)
3. Mandatory	Yes

4. Validation Rules	Currently only 1 value in use ('No'). Future values: 'Yes' with a numeric multiple.
5. Field Kind	Controlled list (dropdown)
6. Valid Options	No Yes (with implied multiple). Future: numeric values like 6, 12, 24.
7. Dependencies	None
8. Default Values	No
9. Duplicates	Duplicates expected.
10. Business Rules	All products currently sell individually ('No'). If set to 'Yes', the Unit field should reflect the multiple (e.g., CTN). Sample: No (2,546 rows).

Sample values: 'No' | 'No' | 'No'

6.6 Commercial Group (Column AM)

Default Price (Column AM, #39)

Distinct values: 166 | Non-null: 2,387 | Null: 159

1. Purpose	The standard retail price in Kuwaiti Dinars (KWD), before any discounts or promotions.
2. Data Type	Number (decimal, 1–3 decimal places)
3. Mandatory	Optional (159 rows have null — price not yet set)
4. Validation Rules	Must be a positive number. KWD uses 3 decimal places (e.g., 1.500).
5. Field Kind	Numeric (currency)
6. Valid Options	N/A
7. Dependencies	None
8. Default Values	None
9. Duplicates	Duplicates expected.
10. Business Rules	Currency: KWD (Kuwaiti Dinar). 1 KWD ≈ 3.25 USD. Sample values: 1.5, 2, 5, 10, 25.

Sample values: '1.5' | '2' | '0.6'

6.7 SEO Group (Columns AN–AR)

SEO Title EN (Column AN, #40)

Distinct values: 1,648 | Non-null: 1,747 | Null: 799

1. Purpose	Search-engine-optimized title for English product pages. Should be keyword-rich and under 60 characters for Google search results.
2. Data Type	String (English)
3. Mandatory	Optional (799 nulls — only set for SEO-prioritized products)
4. Validation Rules	Free text. Recommended length: 50–60 characters.
5. Field Kind	Free text
6. Valid Options	N/A
7. Dependencies	Derived from Name EN (often a shortened version).
8. Default Values	None
9. Duplicates	Duplicates common (1,648 distinct values).
10. Business Rules	Should include primary keyword (product type) + brand + key attribute. Sample: 'Al-Nassim Milk Saucepan'.

Sample values: 'Al-Nassim Milk Saucepan' | 'Al-Nassim Milk Saucepan' | 'Al-Nassim Baby Water Sprayer SH422'

SEO Title AR (Column AO, #41)

Distinct values: 2,535 | Non-null: 2,546 | Null: 0

1. Purpose	Search-engine-optimized title for Arabic product pages.
2. Data Type	String (Arabic)
3. Mandatory	Yes (defaults to Name AR)
4. Validation Rules	Free text. Recommended length: 50–60 characters.
5. Field Kind	Free text (Arabic)
6. Valid Options	N/A
7. Dependencies	Often mirrors Name AR.

8. Default Values	Defaults to Name AR.
9. Duplicates	Duplicates common (2,535 distinct — mirrors Name AR).
10. Business Rules	Should include Arabic primary keyword. Sample: 'كسرولة حليب 14 سم Olina'.

Sample values: 'كسرولة حليب 14 سم Olina' | 'كسرولة حليب 16 سم Olina' | 'مطارة ماء اطفال Al-nassim SH422'

SEO Description EN (Column AP, #42)

Distinct values: 471 | Non-null: 2,546 | Null: 0

1. Purpose	Meta description for English product pages, shown in Google search results.
2. Data Type	String (English)
3. Mandatory	Yes (no nulls — defaults to a generated sentence)
4. Validation Rules	Free text. Recommended length: 150–160 characters.
5. Field Kind	Free text
6. Valid Options	N/A
7. Dependencies	Auto-generated from Brand + Product Type if not explicitly written.
8. Default Values	Defaults to '{Brand} {Product Type}. Available in Kuwait.' format.
9. Duplicates	Duplicates common (471 distinct values).
10. Business Rules	Should include a call-to-action and location. Sample: 'Al-Nassim Pan. Available in Kuwait.'

Sample values: 'Al-Nassim Pan. Available in Kuwait.' | 'Al-Nassim Pan. Available in Kuwait.' | 'Al-Nassim Sprayer. Available in Kuwait.'

SEO Description AR (Column AQ, #43)

Distinct values: 126 | Non-null: 2,546 | Null: 0

1. Purpose	Meta description for Arabic product pages.
2. Data Type	String (Arabic)

3. Mandatory	Yes (no nulls — defaults to a generated sentence)
4. Validation Rules	Free text. Recommended length: 150–160 characters.
5. Field Kind	Free text (Arabic)
6. Valid Options	N/A
7. Dependencies	Auto-generated from Brand AR + Product Type.
8. Default Values	Defaults to '{Brand AR}, {Product Type}' format.
9. Duplicates	Duplicates common (126 distinct values).
10. Business Rules	Sample: 'النسيم الصين', Other'.

Sample values: 'النسيم الصين', Other' | 'النسيم الصين', Other' | 'النسيم الصين', Sprayer'

Search Keywords (Column AR, #44)

Distinct values: 2,474 | Non-null: 2,546 | Null: 0

1. Purpose	Comma/pipe-separated keywords for the e-commerce site's internal search engine. Boosts products in on-site search results.
2. Data Type	String (English, pipe-separated)
3. Mandatory	Yes (no nulls)
4. Validation Rules	Free text. Use pipe ' ' as separator. Recommended: 5–10 keywords per product.
5. Field Kind	Free text (structured)
6. Valid Options	N/A
7. Dependencies	Derived from Brand + Product Family + Product Type + Material.
8. Default Values	Defaults to '{Brand} {Product Family} {Product Type}' format.
9. Duplicates	Duplicates common (2,474 distinct values).
10. Business Rules	Should include synonyms, use cases, and brand variants. Sample: 'Al-Nassim China Pan Cookware'.

Sample values: 'Al-Nassim China | Pan | Cookware' | 'Al-Nassim China | Pan | Cookware' | 'Al-Nassim China | Sprayer | Kitchen Tools | SH422'

6.8 Internal Group (Columns AS–AZ)

Internal Notes (Column AS, #45)

Distinct values: 1 | Non-null: 2 | Null: 2,544

1. Purpose	Free-text notes for internal staff (not shown to customers). Used for supplier details, restock alerts, quality notes.
2. Data Type	String
3. Mandatory	Optional (2,544 nulls — only 2 rows have notes)
4. Validation Rules	Free text.
5. Field Kind	Free text
6. Valid Options	N/A
7. Dependencies	None
8. Default Values	None
9. Duplicates	N/A (essentially unused).
10. Business Rules	Not displayed anywhere customer-facing. Sample: (1 distinct value, 'Other').

Sample values: 'warranty: له كفالة' | 'warranty: له كفالة'

Validation Status (Column AT, #46)

Distinct values: 1 | Non-null: 2,546 | Null: 0

1. Purpose	Indicates whether the product row has passed data-quality validation.
2. Data Type	String
3. Mandatory	Yes
4. Validation Rules	Must come from controlled list. Currently 1 value.
5. Field Kind	Controlled list (dropdown)
6. Valid Options	Validated Pending Review Draft Rejected (future values). Currently all rows are 'Validated'.
7. Dependencies	None

8. Default Values	Validated
9. Duplicates	Duplicates expected.
10. Business Rules	Set to 'Validated' after passing V3–V8 audit pipeline. New products start as 'Draft'. Sample: Validated (2,546 rows).

Sample values: 'Validated' | 'Validated' | 'Validated'

Confidence Score (Column AU, #47)

Distinct values: 10 | Non-null: 2,546 | Null: 0

1. Purpose	Numeric confidence score (0–100) indicating the translation/classification quality of the row.
2. Data Type	Integer (0–100)
3. Mandatory	Yes
4. Validation Rules	Must be an integer between 0 and 100.
5. Field Kind	Numeric (integer)
6. Valid Options	N/A — any integer 0–100
7. Dependencies	Computed from translation audit results.
8. Default Values	94 (default for lexicon-matched translations)
9. Duplicates	Duplicates expected.
10. Business Rules	100 = perfect match, 97 = high-confidence lexicon match, 94 = standard lexicon match, 82–89 = inferred (needs review), 77–79 = low confidence. Sample: 97 (1,337 rows), 94 (978 rows), 82 (117 rows).

Sample values: '94' | '94' | '94'

Pieces (Column AV, #48)

Distinct values: 25 | Non-null: 351 | Null: 2,195

1. Purpose	The number of pieces in the product set (e.g., 6 for a 6-piece knife set).
2. Data Type	Integer (positive)
3. Mandatory	Optional (2,195 nulls — only set for multi-piece

	products)
4. Validation Rules	Must be a positive integer.
5. Field Kind	Numeric (integer)
6. Valid Options	N/A
7. Dependencies	Should appear in Name EN as 'X-Piece' (e.g., '6-Piece').
8. Default Values	None (null = single piece)
9. Duplicates	Duplicates expected.
10. Business Rules	Must match the quantity stated in Name AR. Sample values: 2, 4, 6, 10, 12, 20, 50, 100.

Sample values: '24' | '4' | '8'

Set Count (Column AW, #49)

Distinct values: 10 | Non-null: 65 | Null: 2,481

1. Purpose	The number of items in a coordinated set (e.g., 3 for a 3-piece cookware set). Distinct from Pieces (which counts individual units).
2. Data Type	Integer (positive)
3. Mandatory	Optional (2,481 nulls)
4. Validation Rules	Must be a positive integer.
5. Field Kind	Numeric (integer)
6. Valid Options	N/A
7. Dependencies	Related to Pieces but used for set-type products.
8. Default Values	None
9. Duplicates	Duplicates expected.
10. Business Rules	Used when the product is a set of different items (e.g., 3 pots of different sizes). Sample values: 2, 3, 4, 5, 6.

Sample values: '4' | '4' | '3'

Shape (Column AX, #50)

Distinct values: 3 | Non-null: 110 | Null: 2,436

1. Purpose	The product's shape, for filtering and visual categorization.
2. Data Type	String
3. Mandatory	Optional (2,436 nulls — only set when shape is a buying factor)
4. Validation Rules	Must come from controlled list. Currently 3 values.
5. Field Kind	Controlled list (dropdown)
6. Valid Options	Square Round Rectangular. Future: Oval, Circular.
7. Dependencies	None
8. Default Values	None
9. Duplicates	Duplicates expected.
10. Business Rules	Per lexicon: مربع → 'Square', مدور/دائري → 'Round', مستطيل → 'Rectangular'. Sample: Square (42), Round (38), Rectangular (30).

Sample values: 'Square' | 'Round' | 'Round'

Finish (Column AY, #51)

Distinct values: 3 | Non-null: 12 | Null: 2,534

1. Purpose	The surface finish/coating of the product (e.g., chrome-plated, coated, painted).
2. Data Type	String
3. Mandatory	Optional (2,534 nulls — rarely specified)
4. Validation Rules	Must come from controlled list. Currently 3 values.
5. Field Kind	Controlled list (dropdown)
6. Valid Options	Coated Black Chrome Chrome. Future: Polished, Matte, Powder-coated.
7. Dependencies	None
8. Default Values	None
9. Duplicates	Duplicates expected.
10. Business Rules	Distinct from Material — Finish describes the surface treatment. Sample: Coated (9), Black Chrome (2),

	Chrome (1).
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Sample values: 'Black Chrome' | 'Black Chrome' | 'Chrome'

Additional Information (Column AZ, #52)

Distinct values: 2 | Non-null: 79 | Null: 2,467

1. Purpose	Catch-all field for extra product attributes not captured elsewhere (e.g., 'with Lid', 'Glass Lid').
2. Data Type	String
3. Mandatory	Optional (2,467 nulls)
4. Validation Rules	Free text. Should use lexicon phrases where applicable.
5. Field Kind	Free text (but should use controlled phrases)
6. Valid Options	Common values: 'with Lid', 'Glass Lid'. Future: 'with Wheels', 'Foldable', 'with Handle'.
7. Dependencies	None
8. Default Values	None
9. Duplicates	Duplicates expected.
10. Business Rules	Should be migrated to a structured 'Features' multi-select field in a future schema revision. Sample: 'with Lid' (50), 'Glass Lid' (29).

Sample values: 'Glass Lid' | 'Glass Lid' | 'Glass Lid'

7. Dependent Dropdowns & Validation Rules

This section documents the dependent dropdown relationships that exist or should exist in the Master Catalog. Dependent dropdowns ensure that selecting a parent value filters the available options in a child column, preventing invalid combinations.

7.1 Existing Dependent Dropdowns

The following dependent relationships are conceptually enforced (via the Taxonomy Reference sheet) but are not currently implemented as Excel data-validation dropdowns:

Parent Column	Child Column	Relationship	Lookup Source
Department	Category	Category must be valid under the selected Department	Taxonomy Reference sheet
Category	Subcategory	Subcategory must be valid under the selected Category (or null if category has no subcategories)	Taxonomy Reference sheet
Brand	Brand AR	Brand AR auto-fills from Brand (1:1 mapping)	Brand lookup table (recommended: named range 'BrandLookup')
Brand	Brand Code	Brand Code auto-fills from Brand (1:1 mapping)	Brand lookup table
Color	Color AR	Color AR auto-fills from Color (1:1 mapping)	Color lookup table (recommended: named range 'ColorLookup')
Material	Material AR	Material AR auto-fills from Material (1:1 mapping)	Material lookup table
Capacity	Capacity Unit	Capacity Unit must be set when Capacity is set	Controlled list
Weight	Weight Unit	Weight Unit must be set when Weight is set	Controlled list
Length/Width/	Dimension Unit	Dimension Unit must	Controlled list

Height/Diameter		be set when any dimension is set	
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7.2 Recommended Dependent Dropdowns (Not Yet Implemented)

The following dependent relationships should be added to improve data consistency. They require creating new lookup tables on a hidden 'Lookups' sheet:

Parent Column	Child Column	Why It Matters
Subcategory	Product Family	Each Subcategory should only allow relevant Product Families (e.g., 'Cooking & Utensils' → Cookware, Kitchen Tools, etc.)
Product Family	Product Type	Each Product Family has a defined set of valid Product Types (see Appendix B). Currently 163 Product Types are free-text.
Department	Section Code	Section Code 102 = Houseware, 103 = Supermarket Equipment. Should auto-fill from Department.
Material	Material AR	Currently 19 EN values but only 10 AR values — normalization needed first.

7.3 Data Validation Rules Summary

Rule	Columns Affected	Validation Type
Unique value constraint	Source Row, Product ID, SKU, Barcode	No duplicates allowed
Non-empty constraint	Product ID, SKU, Barcode, Brand, Department, Category, Name AR, Name EN, Short Desc AR/EN, Long Desc EN, Unit, Validation Status, Confidence Score	Must not be null

Numeric constraint	Capacity, Weight, Length, Width, Height, Diameter, Default Price, Confidence Score, Pieces, Set Count	Must be a positive number
Integer constraint	Source Row, Confidence Score, Pieces, Set Count	Must be an integer
Controlled list constraint	Department, Category, Subcategory, Brand, Color, Material, Unit, Capacity Unit, Weight Unit, Dimension Unit, Shape, Finish, Validation Status, Section Code, Country of Origin	Must come from the defined lookup list
Paired constraint	Capacity ↔ Capacity Unit, Weight ↔ Weight Unit, Dimensions ↔ Dimension Unit, Brand ↔ Brand AR ↔ Brand Code, Color ↔ Color AR, Material ↔ Material AR	If one is set, the other must be set
Mutually exclusive	Length/Width vs Diameter	Round products use Diameter; rectangular products use Length+Width. Do not mix.
Format constraint	ND Number (ND-XXXX pattern), Barcode (EAN-13 checksum), Brand Code (3-digit)	Must match the specified format

8. Derived & Auto-Generated Columns

Several columns in the Master Catalog are either auto-generated, auto-derived from other columns, or should be. This section documents the derivation rules.

8.1 Currently Auto-Generated Columns

Column	Derivation Rule	Implemented?
Source Row	Sequential integer assigned when a row is created	Yes — auto-assigned
Short Desc AR	Defaults to Name AR if not explicitly written	Yes — defaulted in V3 audit
Short Desc EN	Defaults to Name EN if not explicitly written	Yes — defaulted in V3 audit
Long Desc EN	Defaults to '{Brand}. {Name EN}.' if not explicitly written	Yes — defaulted in V3 audit
SEO Title AR	Defaults to Name AR	Yes — defaulted
SEO Description EN	Defaults to '{Brand} {Product Type}'. Available in Kuwait.'	Yes — defaulted
SEO Description AR	Defaults to '{Brand AR}, {Product Type}'	Yes — defaulted
Search Keywords	Defaults to '{Brand} {Product Family} {Product Type}'	Yes — defaulted
Validation Status	Defaults to 'Validated' after passing the V3–V8 audit pipeline	Yes — set during audit
Confidence Score	Computed from translation audit results (100=perfect, 97=high, 94=standard, 82=inferred, 77=low)	Yes — set during audit

8.2 Columns That Should Be Auto-Derived (Recommendations)

Column	Should Derive From	Derivation Rule
Brand AR	Brand	VLOOKUP from BrandLookup table: Al-Nassim China → النسيم

		الصين, Local Companies → شركات محلية
Brand Code	Brand	VLOOKUP from BrandLookup table: Al-Nassim China → 017, Local Companies → 006
Color AR	Color	VLOOKUP from ColorLookup table (15-row lookup)
Material AR	Material	VLOOKUP from MaterialLookup table (after normalization)
Section Code	Department	VLOOKUP: Houseware → 102, Supermarket Equipment → 103, Warehouse Equipment → (new code)
Dimension Unit	(any dimension set)	Auto-set to 'cm' when Length/Width/Height/Diameter is first populated
Capacity Unit	Capacity	Auto-set to 'L' (for liquids) or 'kg' (for solids) based on Product Family

8.3 Columns Derived During Translation Audits

The Name EN column is derived from Name AR through the 8-round translation audit pipeline (V3 → V8).

The derivation follows the Al-Nassim Retail Lexicon and these rules:

- **Naming format:** Brand → ND Number → Quantity → Material → Handle → Shape → Product → Accessory → Capacity → Dimensions
- **Lexicon source:** Per the Al-Nassim Retail Lexicon (17 batches + Override Lexicon 18 batches)
- **Preservation rule:** All attributes in Name AR (material, dimensions, capacity, quantity, color, ND number) must be preserved in Name EN
- **Override rule:** Forbidden phrases (Food Container, Sweeper, Shovel, Bin, Flask for براد شاي, etc.) are replaced with lexicon-approved terms

9. Recommendations

Based on the schema analysis, the following recommendations would improve data consistency, reduce manual errors, and streamline catalog management:

9.1 Implement Excel Data Validation Dropdowns

The following columns should have Excel data-validation dropdowns implemented (via Data → Data Validation → List):

Column	Lookup Source	Priority
Brand	Named range 'BrandLookup' on hidden Lookups sheet	High
Department	Named range 'DepartmentList'	High
Category	Named range filtered by Department (INDIRECT formula)	High
Subcategory	Named range filtered by Category	High
Section Code	Named range 'SectionCodeList'	Medium
Product Family	Named range filtered by Subcategory	High
Product Type	Named range filtered by Product Family	High
Color	Named range 'ColorList'	High
Material	Named range 'MaterialList' (after normalization)	High
Unit	Named range 'UnitList'	High
Capacity Unit	Named range 'CapacityUnitList'	Medium
Weight Unit	Named range 'WeightUnitList'	Medium
Dimension Unit	Named range 'DimensionUnitList'	Medium
Shape	Named range 'ShapeList'	Medium
Finish	Named range 'FinishList'	Medium
Country of Origin	Named range 'CountryList' (ISO	Medium

	3166)	
Validation Status	Named range 'ValidationStatusList'	Medium

9.2 Implement Auto-Derivation Formulas

The following columns should be auto-populated via VLOOKUP or INDEX/MATCH formulas (making them read-only):

- Brand AR → VLOOKUP(Brand, BrandLookup, 2, FALSE)
- Brand Code → VLOOKUP(Brand, BrandLookup, 3, FALSE)
- Color AR → VLOOKUP(Color, ColorLookup, 2, FALSE)
- Material AR → VLOOKUP(Material, MaterialLookup, 2, FALSE)
- Section Code → VLOOKUP(Department, DeptSectionLookup, 2, FALSE)

9.3 Normalize Inconsistent Values

Column	Issue	Fix
Material	'Wood' (182) and 'Wooden' (8) coexist	Standardize on 'Wooden' (adjective form per lexicon)
Material	'Iron' (15) used for حديد	Change to 'Steel' per lexicon (حديد → Steel)
Material	'Sheet Steel' (6) used	Merge into 'Steel' or use context-specific 'Saj Griddle'
Capacity Unit	'KG' (5) and 'kg' (31) coexist	Standardize on lowercase 'kg'
Unit	'(1) ربطه' in Arabic	Change to 'BUNDLE'
Country of Origin	Only 'China' in use	Add full ISO 3166 country list to dropdown
Product Type	163 distinct values, some too generic ('Container', 'Set', 'Tool')	Review and split into specific types per the lexicon

9.4 Create a Hidden 'Lookups' Sheet

A hidden 'Lookups' sheet should be created to host all named ranges used by dropdowns and VLOOKUP formulas. This sheet should contain:

- BrandLookup (3 columns: Brand, Brand AR, Brand Code) — 2 rows
- ColorLookup (2 columns: Color EN, Color AR) — 15 rows
- MaterialLookup (2 columns: Material EN, Material AR) — ~15 rows after normalization

- TaxonomyLookup (3 columns: Department, Category, Subcategory) — 21 rows from Taxonomy Reference
- FamilyLookup (2 columns: Subcategory, Product Family) — defines valid Family per Subcategory
- TypeLookup (2 columns: Product Family, Product Type) — defines valid Type per Family (see Appendix B)
- UnitList, CapacityUnitList, WeightUnitList, DimensionUnitList — single-column lists
- ShapeList, FinishList, CountryList, ValidationStatusList — single-column lists

9.5 Populate Empty High-Value Columns

Column	Null Count	Recommendation
Height	2,546 (100% empty)	Populate for tall products (mops, brooms, chairs) — needed for shipping calc
Internal Notes	2,544 (99.9% empty)	Either populate or remove the column
ND Number	155 (6.1% empty)	Assign ND numbers to remaining products for consistency
Country of Origin	1,670 (65.6% empty)	Required for GCC customs — populate for all products
Default Price	159 (6.2% empty)	Populate before publishing to e-commerce
Long Desc AR	1,873 (73.6% empty)	Write long descriptions for premium/featured products
SEO Title EN	799 (31.4% empty)	Write SEO titles for all products (Google ranking impact)

10. Appendix A: Complete Product Type List

The Product Type column currently has 163 distinct values. The complete list, sorted by frequency (descending), is below. Products with very low frequency (1–2 rows) may warrant review for consolidation.

High-Frequency Product Types (≥5 products) — 50 types

Product Type	Product Count
Strainer	158
Container	134
Bucket	133
Wooden Tool	111
Knife	109
Cutter	107
Bag	107
Basket	98
Broom	90
Box	76
Pan	70
Tray	67
Chair	57
Set	56
Cooking Pot	53
Grill	51
Scale	47
Flask	45
Plate	36
Vegetable Slicer	35
Scissors	28
Power Strip	28

Sprayer	26
Serving Bowl	24
Frying Pan	22
Table	22
Cup	21
Juicer	21
Floor Mop	21
Gas Stove	21
Food Storage Container	20
Brush	20
Slotted Spoon	20
Towel	19
Potato Peeler	19
Bottle	18
Cutting Board	18
Fork	17
Peeler	15
Ice Tongs	14
Whisk	14
Gloves	13
Dishwashing Wire	13
Sharpening Steel	12
Casserole	11
Mug	11
Clothesline	11
Rack	11
Coffee Pot	11

Soup Ladle	10
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Low-Frequency Product Types (<5 products) — 0 types

These low-frequency types may indicate specialized products or naming inconsistencies. Review for possible consolidation:

Product Type	Product Count
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11. Appendix B: Product Family → Product Type Mapping

This appendix documents the complete mapping of Product Families to their valid Product Types, based on actual usage in the catalog. This mapping should be used to implement the recommended dependent dropdown (Product Family → Product Type).

Buckets (415 products, 22 types)

Product Type	Count
Bucket	133
Basket	87
Box	64
Strainer	45
Cutter	26
Serving Bowl	11
Mug	11
Juice Straw	7
Water Dispenser	5
Juicer	4
Fruit Juicer	3
Tray	3
Plastic Spoon	3
Food Storage Cover	2
Sprayer	2
Household Ladder	2
Coffee Spoon	2
Measuring Cup	1
Grinder	1
Holder	1
Ice Tongs	1
Masher	1

Cleaning (176 products, 16 types)

Product Type	Count
Broom	89
Brush	18
Floor Mop	15
Cutter	11
Dishwashing Wire	9
Trolley	7
Vacuum Cleaner	6
Dishwashing Sponge	5
Drain Piston Cleaner	4
Mop	3
Juicer	3
Mosquito Repellent	2
Drain Cleaner	1
Sticky Fly Trap	1
Mouse Trap	1
Garbage Bag	1

Cookware (378 products, 21 types)

Product Type	Count
Strainer	111
Pan	70
Cooking Pot	41
Flask	30
Frying Pan	22
Set	16

Tray	13
Casserole	11
Soup Ladle	10
Food Cover	9
Slotted Spoon	7
Serving Bowl	6
Salt Shaker	6
Griddle	6
Cutter	5
Wooden Tool	4
Juicer	3
Funnel	3
Vegetable Chopper	2
Bowl	2
Buffet Warmer	1

Cutlery (242 products, 15 types)

Product Type	Count
Knife	106
Cutter	37
Scissors	28
Fork	16
Wooden Tool	16
Peeler	14
Sharpening Steel	7
Kitchen Cleaver	6

Dining Spoon	3
Dining Fork	3
Bottle	2
Set	1
Cleaver	1
Vegetable Slicer	1
Portable Stove Oven	1

Drinkware (52 products, 4 types)

Product Type	Count
Cup	16
Bottle	16
Flask	15
Pitcher	5

Food Storage (125 products, 1 types)

Product Type	Count
Container	125

Kitchen Tools (578 products, 99 types)

Product Type	Count
Wooden Tool	56
Scale	47
Vegetable Slicer	34
Power Strip	28
Sprayer	24
Gas Stove	21
Food Storage Container	20

Cutter	20
Cutting Board	18
Box	12
Clothesline	11
Rack	11
Coffee Pot	11
Slippers	10
Adhesive Tape	9
Slotted Spoon	8
Whisk	8
Gloves	8
Insecticide	7
Mortar	7
Masher	7
Dining Spoon	6
Toothpick	6
Floor Mop	6
Bowl	6
Potato Peeler	6
Cup	5
Ashtray	5
Juicer	5
Lint Roller	5
Sharpening Steel	5
Hanger	5
Pricing Label Gun	4

Ladle	4
Insect Catcher	4
Dining Cover Curtain	4
Rolling Pin	4
Tray	4
Dishwashing Wire	4
Bath Loofah	4
Household Ladder	4
Clothes Peg	4
Packing Tape	4
Flower Pen	3
Knife	3
Charging Cable	3
Mouse Glue Trap	3
Ice Cream Mold	3
Ice Cream Scoop	3
Cream Decoration Bag	3
Tea Bag	3
Pastry Punch	3
Vegetable Chopper	3
Hand Duster	3
Gas Hose	3
Container	3
Gas Burner Stove	3
Flame Diffuser	3
Drinking Glass Set	2
Ice Tongs	2

Tea Spoon	2
Cleaning Brush	2
Insect Sticky Trap	2
Fan	2
Strainer	2
Cake Decoration Bag	2
Cotton Swab	2
Set	2
Double-Sided Foam Tape	2
Brush	1
Broom	1
Window Squeegee	1
Bathroom Mat	1
Fruit Juicer	1
Roll	1
Pricing Ink Roller	1
Grain Grinder	1
Water Filter	1
Shaving Machine	1
Washing Machine Cover	1
Gas Cover	1
Machine	1
Fire Insulator	1
Binding Thread	1
Pasta Spoon	1
Turner	1
Liquid Dropper	1

Meat Injector	1
Garbage Bag	1
Silicone Spoon	1
Holder	1
Shoe Polish Sponge	1
Clothes Hanger	1
Funnel	1
Shoe Cover	1
Incense	1
Zucchini Corer	1
Display Screen	1
Salt Shaker	1

Linens (19 products, 1 types)

Product Type	Count
Towel	19

Outdoor Cooking (80 products, 7 types)

Product Type	Count
Grill	51
Ice Tongs	11
Tool	7
Gas Heater	5
Plastic Hobby Box	3
Fish Scaler	2
Gas Burner Stove	1

Paper Goods (4 products, 1 types)

Product Type	Count
Tissue	4

Sets (218 products, 6 types)

Product Type	Count
Bag	107
Chair	57
Set	30
Table	22
Dustpan	1
Shelf	1

Storage (124 products, 21 types)

Product Type	Count
Wooden Tool	35
Potato Peeler	13
Basket	11
Cutter	8
Meat Tenderizer	8
Set	7
Container	6
Juicer	6
Whisk	6
Slotted Spoon	5
Vegetable Peeler	5
Fire Insulator	5
Ladle	1

Garlic Press	1
Almond Cracker	1
Peeler	1
Fish Scaler	1
Drum	1
Brush	1
Fork	1
Ice Cream Scoop	1

Tableware (94 products, 4 types)

Product Type	Count
Tray	47
Plate	36
Serving Bowl	7
Plate Rack	4

Uncategorized (41 products, 11 types)

Product Type	Count
Cooking Pot	12
Pressure Cooker	8
Food Storage Bag	5
Gloves	5
Arabic Coffee Pot	3
Hamburger Press	2
Food Storage Cover	2
Bowl	1
Baking Tray	1
Falafel Maker Mold	1

Floor Adhesive	1

12. Appendix C: All Sheets in the Workbook

The V8 workbook contains 63 sheets. They are listed below in physical order (the order they appear in the Excel tab bar).

#	Sheet Name	Size	Purpose
1	Master Catalog	2,548 × 52	Core product table — the only sheet that should be directly edited
2	Translation Audit V3	2,547 × 16	Per-row audit log from V3 (Action: UNCHANGED/CORRECTED/APPLIED/FLAG_REVIEW)
3	Translation Audit Summary V3	92 × 4	V3 roll-up stats and top recurring issues
4	Inferred Classifications	209 × 15	208 products whose classification was inferred (not direct-matched)
5	Classification Summary	34 × 3	V2 classification coverage summary (100% achieved)
6	Validation Report	22 × 4	Taxonomy validation results
7	Taxonomy Reference	22 × 4	Defines the Department → Category → Subcategory hierarchy (21 entries)
8	Translation Report	37 × 6	Translation pipeline summary
9	Translation Fallback Log	197 × 7	Translations that fell back to secondary

			sources
10	Translation Normalization Log	62 × 4	Normalizations applied to translations
11	Translation Manual Review	2 × 7	Translations flagged for manual review
12	Name Rewrite Report	31 × 4	Summary of the V1 name-rewrite pass
13	Name Rewrite Log	196 × 13	196 individual name rewrites (V1)
14	Name Rewrite Report V2	25 × 4	Summary of the V2 phrase-first name-rewrite pass
15	Name Rewrite Log V2	936 × 14	936 individual name rewrites (V2)
16	Audit Report	32 × 4	V2 data-integrity audit summary
17	Audit Change Log	769 × 6	768 individual changes from the V2 audit
18	Translation Quality Audit	19 × 4	Translation quality audit findings
19	Translation Quality Corrections	472 × 13	472 corrections from the quality audit
20	Cross-Validation Audit V3	30 × 4	Cross-validation findings
21	Cross-Validation Discrepancies V3	517 × 5	517 cross-validation discrepancies
22	Attribute Preservation Audit	24 × 4	Attribute preservation findings
23	Attribute Preservation Corrections	102 × 7	102 attribute preservation fixes
24	Retail Terminology	24 × 4	Retail terminology

	Audit		findings
25	Retail Terminology Corrections	56 × 5	56 retail terminology fixes
26	Semantic Translation Audit	32 × 4	Semantic translation findings
27	Semantic Translation Corrections	402 × 6	402 semantic fixes
28	Qualifier Preservation Audit	27 × 4	Qualifier preservation findings
29	Qualifier Preservation Corrections	19 × 5	19 qualifier fixes
30	Cookware Audit	33 × 4	Cookware-specific audit findings
31	Cookware Corrections	95 × 5	95 cookware fixes
32	Attribute Extraction Report	43 × 4	Attribute extraction summary
33	Attribute Extraction Log	2,203 × 4	2,203 attribute extraction records
34	Translation QA Final	31 × 4	Final QA audit findings
35	Translation QA Corrections	1,413 × 6	1,413 final QA corrections
36	Semantic Revalidation	28 × 4	Semantic revalidation findings
37	Semantic Revalidation Corr.	281 × 5	281 revalidation fixes
38	Final Semantic Audit	27 × 4	Final semantic audit findings
39	Final Semantic Corrections	445 × 5	445 final semantic fixes
40	Model Extraction Report	28 × 4	Model code extraction summary

41	Model Extraction Log	145 × 6	145 model extraction records
42	Complete Re-Audit	36 × 4	Complete re-audit findings
43	Complete Re-Audit Corr.	474 × 5	474 complete re-audit fixes
44	Targeted Fixes Audit	35 × 4	Targeted fixes findings
45	Targeted Fixes Corrections	129 × 5	129 targeted fixes
46	Fix13 Audit	22 × 4	Fix batch 13 findings
47	Fix13 Corrections	57 × 5	57 fixes from batch 13
48	Fix15 Audit	24 × 4	Fix batch 15 findings
49	Fix15 Corrections	109 × 5	109 fixes from batch 15
50	Fix11 Audit	20 × 4	Fix batch 11 findings
51	Fix11 Corrections	73 × 5	72 fixes from batch 11
52	Completeness Audit	9 × 4	Translation completeness findings
53	Completeness Corrections	374 × 11	373 completeness fixes
54	Completeness Flagged	69 × 6	68 rows flagged for completeness review
55	Fix10 Audit	19 × 4	Fix batch 10 findings
56	Fix10 Corrections	43 × 5	42 fixes from batch 10
57	V5 Application Summary	74 × 3	V5 application roll-up (226 FLAG_REVIEW rows resolved)
58	V6 Refinement Log	687 × 14	686 rows where Product Type was upgraded from 'Other'
59	V6 Refinement	71 × 4	V6 roll-up stats

	Summary		
60	V7 Override Log	116 × 14	115 rows where forbidden phrases were replaced per Override Lexicon
61	V7 Override Summary	39 × 4	V7 roll-up stats
62	V8 Refinement Log	55 × 11	54 rows with targeted final fixes
63	V8 Refinement Summary	33 × 4	V8 roll-up stats